EXPERIMENT

4.6

Newton's Rings
Wavelength of Sodium Light

Aim

To determine the wavelength of the sodium light by Newton's rings method.

Apparatus

Newton's rings experimental setup. [Two glass plates, convex lens (focal length 1 m), black paper, wooden stand], travelling microscope, sodium lamp.

Newton's rings experimental setup consists of following arrangement. A convex lens L of large focal length (1 m) is placed on an optically plane glass plate G. A black paper is placed under this glass plate. In between the lower surface of the lens L and the upper surface of the glass plate G, there will be a thin layer of air. Another glass plate B is arranged at 45° (to the axis of the lens) just above the lens as shown in Figure 1. Usually, these arrangements are made on a wooden stand.

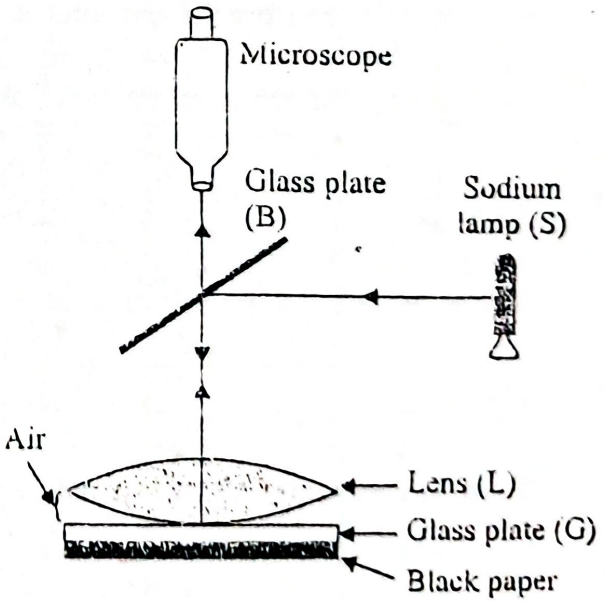


Figure 1 Newton's rings experimental setup.

Sodium vapour lamp S is arranged (at a distance of 1 m from the glass plate B) renders a horizontal parallel beam of light to the glass plate B, and then reflects a part of it to the lens. Interference take places between the light beams reflected from the lower surface of lens L and the upper surface of glass plate G.

The interference fringes (Figure 2) can be viewed with the microscope of a travelling microscope arranged vertically above the glass plate B. The fringe system consists of alternate bright and dark circular rings with a dark spot at the centre.

Note: Circular rings can also be seen with the naked eyes when looked through the glass plate B at the point where the convex lens L touches the glass plate G.

Principle

According to the theory of Newton's rings, the diameter of the nth ring is given by

$$D_n = \sqrt{4Rn\lambda}$$

Then

$$D^2_{n+m} - D^2_n = 4R(n + m)\lambda - 4Rn\lambda$$
$$= 4Rm\lambda$$

or the wavelength of the sodium light

$$\lambda = \frac{D^2_{n+m} - D^2_n}{4mR}$$

where R is the radius of curvature of the convex lens n and m are the order of the circular rings. By Boy's method, the radius of curvature of the lens

$$R = \frac{fd}{f - d}$$

where f is the focal length which can be found by uv method $\left(f = \frac{uv}{u + v}\right)$ and d is the distance to the image from the lens by Boy's method.

Procedure

Note the least count of the travelling microscope (horizontal scale).

Focus the microscope (by adjusting the eyepiece and its tangential screws) such that the ring system can be clearly seen. By working the horizontal movement of the travelling microscope, make sure that about 30 rings can be viewed on both sides of the ring system.

Adjust the screws (both horizontal and vertical movement screws) of the travelling microscope and bring the cross wires to the centre of the rings. By working horizontal movement screws, bring the vertical cross wire tangential to the 20th ring on the left. Find the horizontal scale reading (MSR + VSR × LC) (which gives the position of the 20th ring on the left). Then move towards 18th, 16th, etc. rings and the corresponding horizontal scale readings are noted (a). This is repeated to the right side of the ring system up to the 20th ring (b).

Note: The horizontal screws should be turned in one direction only to avoid back slash error.

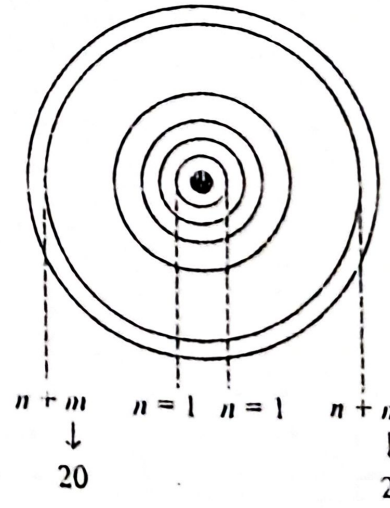


Figure 2 Schematic of Newton's rings.

Find the diameter of the 20th, 18th, etc. rings by finding the difference between the left and right horizontal scale readings ($a \sim b$), and then, $D_{n+m}^2 - D_n^2$. From this, the mean value for $D_{n+m}^2 - D_n^2$ for $m = 10$ is calculated.

Find the focal length f of the lens by uv method. Then, by Boy's method, calculate the radius of curvature by finding the distance between the lens and the image (d) as

$$R = \frac{fd}{f - d}$$

Then the focal length of the sodium light

$$\lambda = \frac{D_{n+m}^2 - D_n^2}{4mR}$$

Observations

To find the least count of the travelling microscope

Value of 1 main scale division = ... mm
No. of divisions on the vernier = ...
Least count

$$LC = \frac{1 \text{ MSD}}{\text{No. of divisions on the vernier}} = \ldots \text{ mm}$$

$$= \ldots \text{ cm}$$

Order n	Travelling microscope reading						Diameter of the nth ring $D_n(a \sim b)$ (cm)	D_n^2 (cm^2)	$D_{n+m}^2 - D_n^2$ (cm^2)
	Left of the ring system			Right of the ring system					
	MSR (cm)	VSR	MSR + VSR × LC (a) (cm)	MSR (cm)	VSR	MSR + VSR × LC (b) (cm)			
20								D_{20}^2	$D_{20}^2 - D_{10}^2$
18								D_{18}^2	$D_{18}^2 - D_8^2$
16									$D_{16}^2 - D_6^2$
14									$D_{14}^2 - D_4^2$
12								D_{12}^2	$D_{12}^2 - D_2^2$
10									
8									
6									
4									
2									

Mean value of $D^2_{n+m} - D^2_n$ = ... cm^2 = ... m^2

Value of order m = 10

Focal length of the convex lens[1]

S. no	Distance to the lens from the		Focal length $f = \frac{uv}{u+v}$ (cm)
	Wire gauze u(cm)	Screen v(cm)	
1.			
2.			
3.			
4.			
5.			

Focal length of the lens f = ... m

Radius of curvature of the lens[2]

Distance to the lens from the object d(cm)				Radius of curvature $R = \frac{fd}{f-d}$ (cm)
1	2	3	Mean d' (cm)	

Radius of curvature of the lens R = ... m

Wavelength of the sodium light $\lambda = \frac{D^2_{n+m} - D^2_n}{4mR}$ = .. m = ... nm

Result

Wavelength of the sodium light = ... nm.

Question

Find the wavelength of sodium light using Newton's rings method.

Notes:

1. Focal length of the lens can be found by using a mirror.
2. The radius of curvature of the lower surface of the lens which is in contact with the glass plate.